Dashboard Design

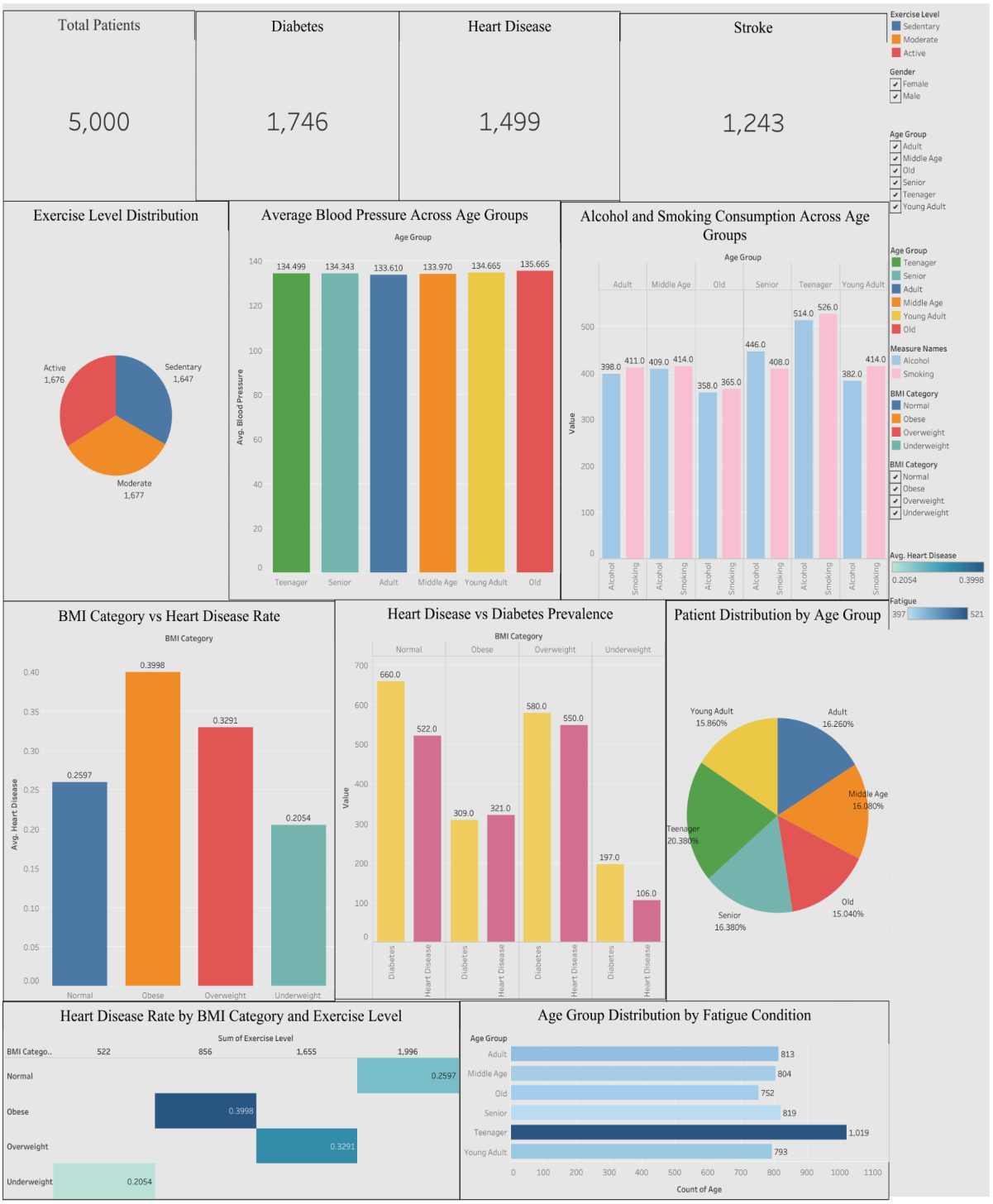
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Team ID	EN25CS3090269
Project Name	Smart Healthcare Analytics
Maximum Marks	5 Marks

Creating an effective dashboard involves thoughtful design to ensure that the presented information is clear, relevant, and easily understandable for the intended audience. Here are some key principles and best practices for dashboard design

Activity 1: Interactive and visually appealing dashboards

Creating interactive and visually appealing dashboards involves a combination of thoughtful design, effective use of visual elements, and the incorporation of interactive features. Here are some tips to help you design dashboards that are both visually appealing and engaging for users so take care of below points

- Clear and Intuitive Layout
- Use Appropriate Visualizations
- Colour and Theming
- Interactive Filters and Slicers
- Drill-Down Capabilities
- Responsive Design
- Custom Visuals and Icons
- Use of Infographics


Total Patients
5,000
Diabetes
1,746
Heart Disease
1,499
Stroke
1,243
Exercise Level
Sedentary
Moderate
Active
Gender
Female
Male
Age Group
Adult
Middle Age
Old
Senior
Teenager
Young Adult
Exercise Level Distribution
Active 1,676
Sedentary 1,647
Moderate 1,677
Average Blood Pressure Across Age Groups
Age Group
Avg. Blood Pressure
134.499
134.343
133.610
133.970
134.665
135.665
Teenager
Senior
Adult
Middle Age
Young Adult
Old
Alcohol and Smoking Consumption Across Age Groups
Age Group
Value
Adult
Middle Age
Old
Senior
Teenager
Young Adult
398.0
411.0
409.0
414.0
358.0
365.0
446.0
408.0
514.0
526.0
382.0
414.0
Alcohol
Smoking
Measure Names
Alcohol
Smoking
BMI Category
Normal
Obese
Overweight
Underweight
Avg. Heart Disease
0.2054
0.3998
Fatigue
397
521
BMI Category vs Heart Disease Rate
BMI Category
Avg. Heart Disease
0.2597
0.3998
0.3291
0.2054
Normal
Obese
Overweight
Underweight
Heart Disease vs Diabetes Prevalence
BMI Category
Value
660.0
522.0
309.0
321.0
580.0
550.0
197.0
106.0
Diabetes
Heart Disease
Patient Distribution by Age Group
Young Adult 15.860%
Adult 16.260%
Middle Age 16.080%
Teenager 20.380%
Old 15.040%
Senior 16.380%
Heart Disease Rate by BMI Category and Exercise Level
Sum of Exercise Level
BMI Catego..
522
856
1,655
1,996
Normal
0.2597
Obese
0.3998
Overweight
0.3291
Underweight
0.2054
Age Group Distribution by Fatigue Condition
Age Group
Adult
813
Middle Age
804
Old
752
Senior
819
Teenager
1,019
Young Adult
793
Count of Age

Here are five potential outcomes from the dashboard provided:

- **Overall Patient and Disease Overview:** The dashboard represents 5,000 patients, with 1,746 recorded diabetes cases, 1,499 heart disease cases, and 1,243 stroke cases. These KPI cards provide a quick overview of the dataset and its recorded disease indicators.

- **Exercise Level Distribution:** The dataset contains 1,677 patients in the moderate exercise category, 1,676 in the active category, and 1,647 in the sedentary category. The distribution is relatively balanced across the three exercise levels.

- **Average Blood Pressure Across Age Groups:** The average blood pressure values range from approximately 133.61 to 135.67 across the six age groups. The Old group has the highest average (135.665), while the Adult group has the lowest (133.610).

- **Heart Disease Rate by BMI Category:** The Obese category has the highest average heart disease indicator at 0.3998 (39.98%), followed by Overweight at 0.3291 (32.91%), Normal at 0.2597 (25.97%), and Underweight at 0.2054 (20.54%).

- **Alcohol and Smoking Consumption Across Age Groups:** The chart compares the displayed alcohol and smoking values across age groups. The highest values appear in the Teenager group, at 514 for alcohol and 526 for smoking, while the lowest alcohol value is 358 in the Old group.

- **Heart Disease and Diabetes Prevalence:** The comparison chart shows higher diabetes counts than heart disease counts in each BMI category. The Normal category has 660 diabetes cases and 522 heart disease cases, while the Obese category has 309 diabetes cases and 321 heart disease cases.

- **Patient Distribution by Age Group:** Teenagers represent the largest displayed age group, with 20.38% of patients, followed by Seniors at 16.38%, Young Adults at 15.86%, Adults at 16.26%, Middle Age at 16.08%, and the Old group at 15.04%.

- **Age Group Distribution by Fatigue Condition:** The displayed counts vary across age groups. Teenagers have the highest count at 1,019, followed by Seniors at 819, Adults at 813, Middle Age at 804, Young Adults at 793, and the Old group at 752.

- **Interactive Dashboard Filtering:** The dashboard includes filters for Exercise Level, Gender, Age Group, and BMI Category. These allow users to explore subsets of the dataset and compare the displayed healthcare indicators.

Activity 2

Publish Dashboard on Tableau Public and Paste the Dashboard Public link below:

https://public.tableau.com/app/profile/sherin.kuriakose/viz/Health360Project-FINAL/Dashboard1